

# Generation Interconnection Request System Impact Study Report

## For Queue Project Q17-0900

Wabash 500 kV  
October 19, 2018

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0900. The Interconnection Customer proposes a 249.0 MW wind facility interconnecting at the Wabash 500 kV substation in the Indiana area. The request was received on December 09, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 408 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$29,355,000.

### **3. Network Impacts**

#### **3.1 Generator Deliverability and Single Contingency Analysis**

The MYRTLE PT to GEORGE NEAL S 230 kV is overloaded to 126.5% of its 670 MVA emergency rating for the loss of the SWAN VALLEY to TREGO 500 kV. The project contributes 8.4% DFAX (21.0 MW) to this overload.

For the loss of SWAN VALLEY to TREGO 500 kV, the ZONA LIBRE 6007 to GEORGE NEAL S 230 kV line loads to 125.4% of its 694 MVA rating. Project DFAX: 8.4%.

For the loss of TREGO to SWAN VALLEY 500 kV, the ZONA LIBRE 6007 to KAMPSVILLE 230 kV Line loads to 124.4% of its 719 MVA rating. Project DFAX: 8.4%.

The MORENO TAP to KAMPSVILLE 230 kV Line is overloaded to 123.8% of its 733 MVA emergency rating for the loss of the SWAN VALLEY to TREGO 500 kV. The project contributes 8.4% DFAX (21.0 MW) to this overload.

EASTOWN ROAD to MORENO TAP 230 kV L: 122.9% loading (758 MVA) following the outage of the TREGO to SWAN VLY 500 kV line; DFAX = 8.4%.

For the loss of SWAN VLY to MUSKOGEE 500 kV, the DEAN FOREST to EASTOWN ROAD 230 kV loads to 114.5% of its 643 MVA rating. Project DFAX: 10.6%.

DEAN FOREST to MUSKOGEE 230 kV: 113.7% loading (659 MVA) following the outage of the MUSKOGEE to SWAN VALLEY 500 kV L; DFAX = 10.6%.

The TIMBER SWITCH to MUSKOGEE 230 kV line is overloaded to 107.7% of its 590 MVA emergency rating for the loss of the SWAN VLY to TREGO 500 kV. The project contributes 7.0% DFAX (17.4 MW) to this overload.

LONETREE to SABANERA SUBSTATION 500 kV: 105.6% loading (2,866 MVA) following the outage of the MUSKOGEE to MONTPELIER ELECTRIC GENERATING STA 500 kV; DFAX = 43.5%.

For the loss of SWAN VLY to TREGO 500 kV, the SWAN VALLEY 500 kV/230 kV Autotransformer loads to 104.7% of its 1,347 MVA rating. Project DFAX: 16.5%.

W MELROSE to TIMBER SWITCH 230 kV: 104.7% loading (639 MVA) following the outage of the SWAN VALLEY to TREGO SUBSTATION 500 kV; DFAX = 6.9%.

The MUSKOGEE 500/230 kV Auto is overloaded to 104.5% of its 1,173 MVA emergency rating for the loss of the SWAN VALLEY to MUSKOGEE 500 kV L. The project contributes 19.5% DFAX (48.5 MW) to this overload.

The TRUAN to LOOKINGGLASS 230 kV is overloaded to 103.9% of its 575 MVA emergency rating for the loss of the TREGO to BEAVER SUBSTATION 500 kV. The project contributes 8.0% DFAX (20.0 MW) to this overload.

W MELROSE to TREGO 230 kV: 103.6% loading (524 MVA) following the outage of the TREGO to SWAN VALLEY 500 kV L; DFAX = 6.8%.

For the loss of EAST METHUEN to LONETREE 500 kV line, the ST. LAWRENCE to JAMES DE YOUNG 230 kV loads to 101.3% of its 1,026 MVA rating. Project DFAX: 12.6%.

MONTPELIER ELECTRIC GENERATING STA 500 kV/230 kV XFMR: 101.1% loading (1,272 MVA) following the outage of the MONTPELIER ELECTRIC GENERATING STA to WABASH 500 kV; DFAX = 13.0%.

The LONETREE to E METHUEN 500 kV L is overloaded to 100.8% of its 2,749 MVA emergency rating for the loss of the MUSKOGEE to MONTPELIER ELECTRIC GENERATING STATION 500 kV Line. The project contributes 40.4% DFAX (100.6 MW) to this overload.

CANAL ROAD to NEWPORT TIE 230 kV: 100.7% loading (1,013 MVA) following the outage of the BEAVER to TREGO 500 kV; DFAX = 9.9%.

For the loss of LONETREE to SABANERA 500 kV L, the SWAN VLY to MUSKOGEE 500 kV loads to 100.3% of its 2,943 MVA rating. Project DFAX: 43.1%.

### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

## **4. Required Network Upgrades and Cost Estimates**

| Facility                                     | Required Upgrade                            | Cost Estimate | Allocation to Project |
|----------------------------------------------|---------------------------------------------|---------------|-----------------------|
| GEORGE NEAL S to MYRTLE PT 230 kV            | Rebuild line with higher-capacity conductor | \$23,138,000  | 4% (\$1,045,000)      |
| ZONA LIBRE 6007 to GEORGE NEAL S 230 kV line | Rebuild line with higher-capacity conductor | \$43,463,000  | 4% (\$1,963,000)      |
| ZONA LIBRE 6007 to KAMPSVILLE 230 kV Line    | Rebuild line with higher-capacity conductor | \$32,779,000  | 4% (\$1,480,000)      |
| MORENO TAP to KAMPSVILLE 230 kV Line         | Rebuild line with higher-capacity conductor | \$57,450,000  | 4% (\$2,594,000)      |
| EASTOWN ROAD to MORENO TAP 230 kV L          | Rebuild line with higher-capacity conductor | \$46,925,000  | 4% (\$2,119,000)      |
| DEAN FOREST to EASTOWN ROAD 230 kV           | Rebuild line with higher-capacity conductor | \$21,693,000  | 6% (\$1,414,000)      |
| DEAN FOREST to MUSKOGEE 230 kV               | Rebuild line with higher-capacity conductor | \$45,059,000  | 6% (\$2,938,000)      |
| TIMBER SWITCH to MUSKOGEE 230 kV line        | Reconductor line                            | \$5,362,000   | 6% (\$305,000)        |
| LONETREE to SABANERA 500 kV line             | Reconductor line                            | \$58,361,000  | 4% (\$2,476,000)      |
| SWAN VALLEY 500 kV/230 kV Autotransformer    | Replace transformer with higher-rated unit  | \$37,376,000  | 5% (\$2,027,000)      |
| W MELROSE to TIMBER SWITCH 230 kV            | Reconductor line                            | \$62,309,000  | 6% (\$3,579,000)      |
| MUSKOGEE 500/230kV TX                        | Replace transformer with higher-rated unit  | \$23,207,000  | 9% (\$2,067,000)      |

| Facility                                              | Required Upgrade                           | Cost Estimate | Allocation to Project |
|-------------------------------------------------------|--------------------------------------------|---------------|-----------------------|
| TRUMAN to LOOKINGGLASS 230 kV                         | Reconductor line                           | \$8,507,000   | 2% (\$156,000)        |
| W MELROSE to TREGO 230 kV                             | Reconductor line                           | \$3,511,000   | 6% (\$199,000)        |
| ST. LAWRENCE to JAMES DE YOUNG 230 kV                 | Reconductor line                           | \$18,283,000  | 2% (\$464,000)        |
| MONTPELIER ELECTRIC GENERATING STA 500 kV/230 kV XFMR | Replace transformer with higher-rated unit | \$29,146,000  | 2% (\$693,000)        |
| LONETREE to E METHUEN 500 kV L                        | Reconductor line                           | \$28,157,000  | 4% (\$1,190,000)      |
| CANAL ROAD to NEWPORT TIE 230 kV                      | Reconductor line                           | \$5,157,000   | 2% (\$101,000)        |
| SWAN VLY to MUSKOGEE 500 kV                           | Reconductor line                           | \$59,035,000  | 4% (\$2,545,000)      |
|                                                       | Total allocated to project                 |               | \$29,355,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.